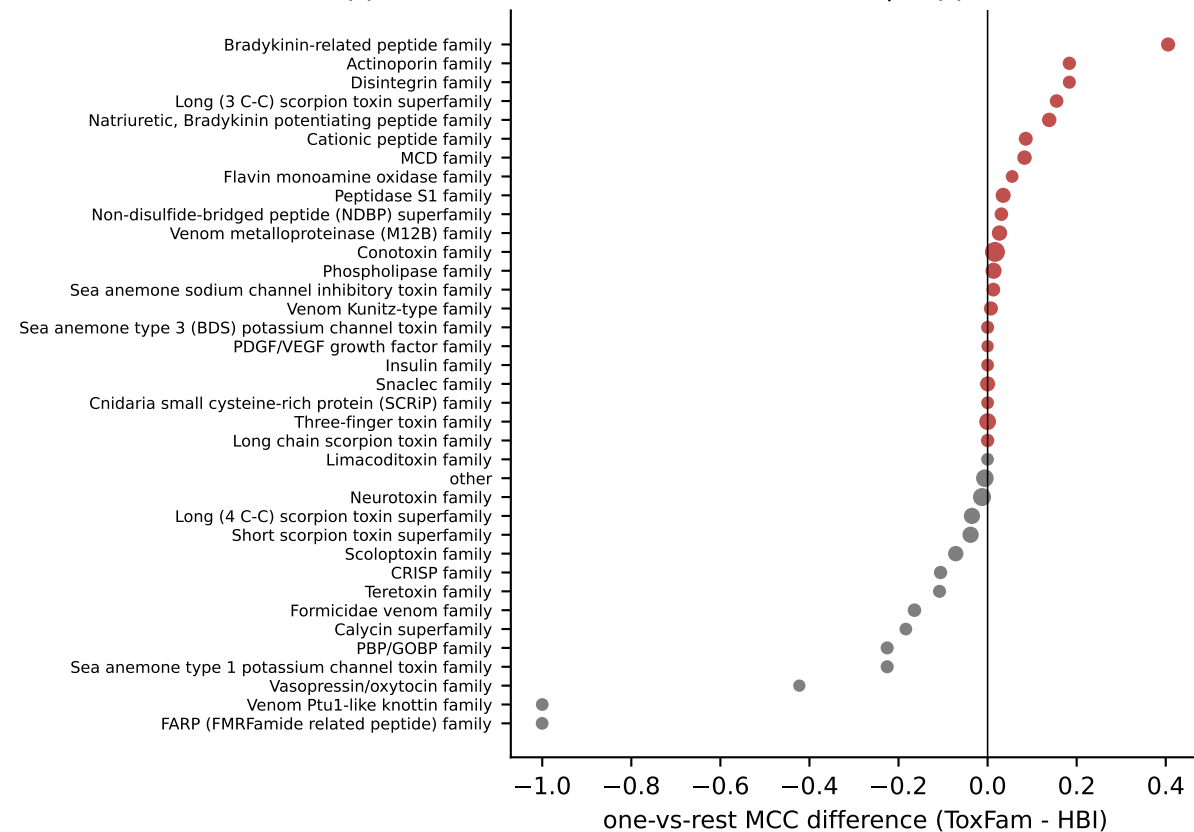


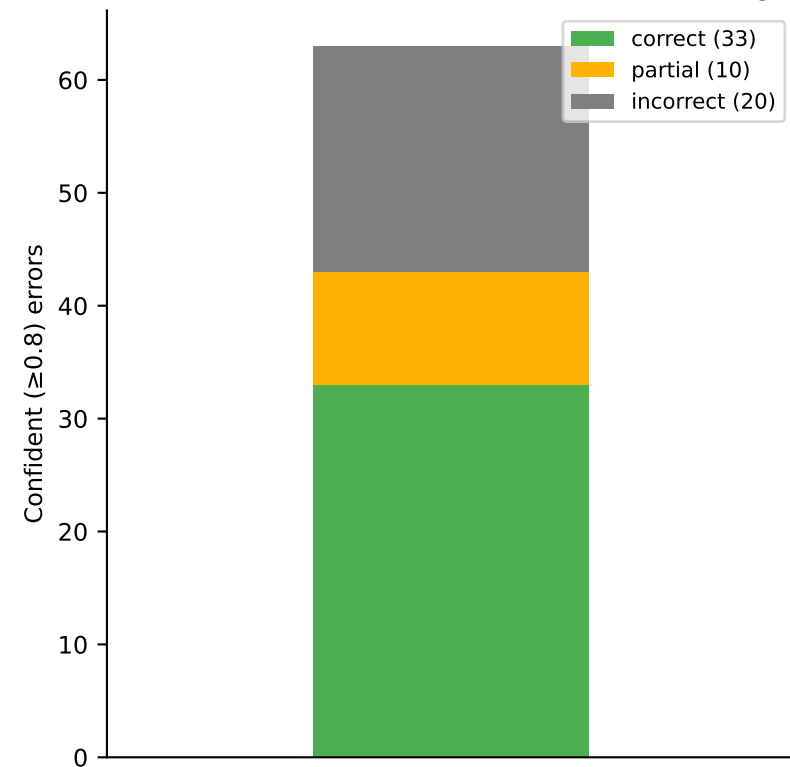
A. Per-family MCC (marker α support)

support>5: ToxFam 0.880 vs HBI 0.842 | support $\leq$ 5 (n=58): 0.723 vs 0.865



B. Adjudicated confident errors (n=63)

43/63 model-vindicated; 38 candidate ToxProt gaps



e.g. Q4VRI5 (PLA2), B2ZBB8 (Kunitz) — labelled nontox, expert verdict toxin